

Applied Human Computer Interaction

Assignment#02



23K-2001

BCS-6J

Q1)

A hot spot is the exact active area of a graphical object where a pointer click is registered. It is important because users rely on visual cues to interact correctly.

- **Accuracy:** The clickable area should match the visual shape (example: a full button should be clickable, not just the text).
- **Predictability:** Users expect the action to occur exactly where they click.
- **Usability & consistency:** Similar UI elements should have similar hot spot behavior to avoid confusion.

Example: In a food delivery app, if the whole food card is clickable instead of just the title text, users can select items faster and with fewer errors.

Q2)

Keyboard accelerators are shortcut key combinations that allow users to execute commands faster than using menus. In Ali's case, shortcuts like Ctrl+S and Ctrl+C reduce interaction time.

- They improve efficiency by reducing mouse movement.
- They support expert users by enabling faster interaction.
- They reduce physical effort during repetitive tasks.

One limitation is that shortcuts must be memorized, which can be difficult for new users.

Q3)

Engelbart's idea of augmenting human intellect meant computers should enhance human thinking rather than just perform calculations. The delay in recognition happened because early systems were expensive, complex, and people were not ready to adopt interactive computing concepts.

Engelbart concluded that the key entity behind intellectual improvement is the **human-tool-method system**, meaning effective problem solving comes from the combined improvement of:

- Human skills
- Tools (computers)
- Methods (ways of working)

Q4)

A full-page word processor mostly follows Shneiderman's direct manipulation ideas because users can see the document, directly select text, and instantly see changes.

However, some features break the pen-paper metaphor:

- Undo/redo (not possible in real writing)
- Copy-paste (no physical equivalent)
- Spell check and auto-formatting

These features improve efficiency but move interaction beyond the traditional writing model.

Q5)

Cultural probes are useful in sensitive research because they let participants share personal data privately without direct observation.

Example: Studying gym habits through self-recorded logs instead of researcher observation.

Difference from ethnography:

- Ethnography → direct observation
- Cultural probes → self-reporting by participants

Cultural probe package (Physical health habits 20–30 age group):

- **7-day health diary:** Record sleep, meals, water intake, and exercise.
- **Photo task:** Take 3 photos of healthy habits and 3 of unhealthy habits during the week.
- **Habit challenge card:** Ask participants to try one small healthy change and note difficulty.
- **Mood & energy tracker:** Simple daily rating (Low / Medium / High) to relate health with feelings.

This package encourages natural responses while keeping participants comfortable sharing personal health behavior.